New L J Institute of Engineering & Technology

SEM: 1st – CSE/CSE (AIML)/DS

Sub: Mathematics-I

Sub Code:BE01R00041

Unit 2: Single variable Calculus (Differentiation)

Weightage: 20%

Topics:

- **Chap:1 (5%)**

  Indeterminate Forms and L'Hôspital's Rule.

- **Chap: 2 (15%)**

  Taylor's and Maclaurin's theorem for a function of one variable,

  Taylor's and Maclaurin's series of a function using statement of the theorems,

  Extreme values of functions

Module - 2:
Taylor's and Maclaurin's series

	TAYLOR SERIES
	Taylor's Theorem : Let f be a real-valued function defined on (a, b).Suppose f has derivatives of all orders upto and including n+1 in the interval(a, b).Let X_0 be any point of the interval (a, b).then for any X in (a, b), $$f(x) = f(x_0) + \frac{x - x_0}{1!} f'(x_0) + \cdots + \frac{(x - x_0)^n}{n!} f^{(n)}(x_0) + \frac{(x - x_0)^{n+1}}{(n+1)!} f^{(n+1)}(c)$$ Where c is a point between x_0 and x. the expression is called Taylor's expansion of $f(x)$ at x_0. **Here,**

$$R_{n+1}(x) = \frac{(x-x_0)^{n+1}}{(n+1)!} f^{(n+1)}(c),$$

we say that the Taylor's series of $f(x)$ at x,converges to the given function for all those x for which $\lim_{n\to\infty} R_{n+1}(x) = 0$.

Taylor's Series:

Statement: If **f(x+h)** is a function of h which can be expanded in to a convergent series of positive ascending powers of $\boldsymbol{h}$ then

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \cdots + \frac{h^n}{n!}f^n(x) + \cdots$$

NOTE: Remember this expansion also use to evaluate examples

(1)*Replacing x by h and h by x, we get expansion in power of x*

$$f(x+h) = f(h) + xf'(h) + \frac{x^2}{2!}f''(h) + \frac{x^3}{3!}f'''(h) + \cdots + \frac{x^n}{n!}f^n(h) + \cdots$$

(2)*Putting* $x = a$ *and* $h = x - a$,

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \cdots + \frac{(x-a)^n}{n!}f^n(a) + \cdots$$

NUMERICAL QUESTIONS

1 Expand e^x in powers of (x – 1) by Taylor's series.

Solution:

By Taylor's series

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \cdots + \frac{(x-a)^n}{n!}f^n(a) + \cdots$$

Let $f(x) = e^x$ and $a = 1$

$f(x) = e^x \Rightarrow f(1) = e$

$f'(x) = e^x \Rightarrow f'(1) = e$

$f''(x) = e^x \Rightarrow f''(1) = e$

$f'''(x) = e^x \Rightarrow f'''(1) = e$

	$f''''(x) = e^x \Rightarrow f''''(1) = e$ Substituting the values in series $e^x = f(1) + (x-1)f'(1) + \frac{(x-1)^2}{2!}f''(1) + \frac{(x-1)^3}{3!}f'''(1) + \cdots + \frac{(x-a)^n}{n!}f^n(a) + \cdots$ $= e + (x-1)e + \frac{(x-1)^2}{2!}e + \frac{(x-1)^3}{3!}e + \frac{(x-1)^4}{4!}e + \cdots$ $= e[1 + (x-1) + \frac{(x-1)^2}{2!} + \frac{(x-1)^3}{3!} + \frac{(x-1)^4}{4!} + \cdots]$
2	Find the Taylor series expansion of $f(x) = x^3 - 2x + 4$, $a = 2$ OR Expand $f(x) = x^3 - 2x + 4$ in powers of $(x-2)$. **Solution:** By Taylor's series $f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \cdots + \frac{(x-a)^n}{n!}f^n(a) + \cdots$ Let $f(x) = x^3 - 2x + 4$ and $a = 2$ $f(x) = x^3 - 2x + 4 \Rightarrow f(2) = 8$ $f'(x) = 3x^2 - 2 \Rightarrow f'(2) = 10$ $f''(x) = 6x \Rightarrow f''(2) = 12$ $f'''(x) = 6 \Rightarrow f'''(2) = 6$ $f''''(x) = 0 \Rightarrow f''''(2) = 0$ Substituting the values in series $x^3 - 2x + 4 = f(2) + (x-2)f'(2) + \frac{(x-2)^2}{2!}f''(2) + \frac{(x-2)^3}{3!}f'''(2) + \cdots + \frac{(x-2)^n}{n!}f^n(2) + \cdots$ $= 8 + (x-2)10 + \frac{(x-2)^2}{2!}12 + \frac{(x-2)^3}{3!}6 + \frac{(x-2)^4}{4!}0 + \cdots$ $= 8 + 10(x-2) + 6(x-2)^2 + (x-2)^3$
3	Expand the polynomial $f(x) = x^5 + 2x^4 - x^2 + x + 1$ in power of $x + 1$. **Solution:** By Taylor's series

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \cdots + \frac{(x-a)^n}{n!}f^n(a) + \cdots$$

Let $f(x) = x^5 + 2x^4 - x^2 + x + 1$ and $a = -1$

$f(x) = x^5 + 2x^4 - x^2 + x + 1 \Rightarrow f(-1) = 0$

$f'(x) = 5x^4 + 8x^3 - 2x + 1 \Rightarrow f'(-1) = 0$

$f''(x) = 20x^3 + 24x^2 - 2 \Rightarrow f''(-1) = 2$

$f'''(x) = 60x^2 + 48x \Rightarrow f'''(-1) = 12$

$f''''(x) = 120x + 48 \Rightarrow f''''(-1) == -72$

$f^V(x) = 120 \Rightarrow f^V(-1) = 120$

Substituting the values in series

$x^5 + 2x^4 - x^2 + x + 1$

$$= f(-1) + (x+1)f'(-1) + \frac{(x+1)^2}{2!}f''(-1) + \frac{(x+1)^3}{3!}f'''(-1) + \cdots + \frac{(x+1)^n}{n!}f^n(-1) + \cdots$$

$$= 0 + (x+1)0 + \frac{(x+1)^2}{2!}2 + \frac{(x+1)^3}{3!}12 + \frac{(x+1)^4}{4!}(-72) + \frac{(x+1)^5}{5!}(120) \ldots$$

$$= (x+1)^2 + 2(x+1)^3 - 3(x+1)^4 + (x+1)^5$$

4 Express $5 + 4(x-1)^2 - 3(x-1)^3 + (x-1)^4$ in ascending powers of x

Solution:

By Taylor's series

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \cdots + \frac{(x-a)^n}{n!}f^n(a) + \cdots$$

Let $f(x) = 5 + 4(x-1)^2 - 3(x-1)^3 + (x-1)^4$ and $a = 0$

$f(x) = 5 + 4(x-1)^2 - 3(x-1)^3 + (x-1)^4 \Rightarrow f(0) = 13$

$f'(x) = 8(x-1) - 9(x-1)^2 + 4(x-1)^3 \Rightarrow f'(0) = -21$

$f''(x) = 8 - 18(x-1) + 12(x-1)^2 \Rightarrow f''(0) = 38$

$f'''(x) = -18 + 24(x-1) \Rightarrow f'''(0) = -42$

$f''''(x) = 24 \Rightarrow f''''(0) = 24$

Exploring Emerging Technologies

Substituting the values in series

$$5+4(x-1)^2-3(x-1)^3+(x-1)^4$$

$$=f(0)+(x-0)f'(0)+\frac{(x-0)^2}{2!}f''(0)+\frac{(x-0)^3}{3!}f'''(0)+\cdots$$

$$+\frac{(x-0)^n}{n!}f^n(0)+\cdots$$

$$=13+x(-21)+\frac{(x)^2}{2!}38+\frac{(x)^3}{3!}(-42)+\frac{(x)^4}{4!}(24)$$

$$=13-21x+19x^2-7x^3+x^4$$

5 Expand $\sin(x+\frac{\pi}{4})$ in powers of x. Hence find the value of $\sin 44°$ and $\sin 46°$

Solution:

By Taylor's series

$$f(x+h)=f(h)+xf'(h)+\frac{x^2}{2!}f''(h)+\frac{x^3}{3!}f'''(h)+\cdots+\frac{x^n}{n!}f^n(h)+\cdots$$

Suppose $\sin\left(x+\frac{\pi}{4}\right)=f(x+h)\Rightarrow f(x)=\sin x$ and $h=45°=\frac{\pi}{4}$

Now

$f(x)=\sin x \qquad \Rightarrow f(h)=f\left(\frac{\pi}{4}\right)=\sin\left(\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}}$

$f'(x)=\cos x \qquad \Rightarrow f'\left(\frac{\pi}{4}\right)=\cos\left(\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}}$

$f''(x)=-\sin x \qquad \Rightarrow f''\left(\frac{\pi}{4}\right)=-\sin\left(\frac{\pi}{4}\right)=-\frac{1}{\sqrt{2}}$

$f'''(x)=-\cos x \qquad \Rightarrow f'''(\frac{\pi}{4})=-\cos\left(\frac{\pi}{4}\right)=-\frac{1}{\sqrt{2}}$

$f''''(x)=\sin x \qquad \Rightarrow f''''\left(\frac{\pi}{4}\right)=\sin\left(\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}}$

Substituting the values in series

$$f(x+h)=f(h)+xf'(h)+\frac{x^2}{2!}f''(h)+\frac{x^3}{3!}f'''(h)+\cdots+\frac{x^n}{n!}f^n(h)+\cdots$$

$$\sin\left(x+\frac{\pi}{4}\right)=f\left(\frac{\pi}{4}\right)+xf'\left(\frac{\pi}{4}\right)+\frac{x^2}{2!}f''\left(\frac{\pi}{4}\right)+\frac{x^3}{3!}f'''\left(\frac{\pi}{4}\right)+\cdots+\frac{x^n}{n!}f^n\left(\frac{\pi}{4}\right)+\cdots$$

$$=\frac{1}{\sqrt{2}}+x\frac{1}{\sqrt{2}}+\frac{x^2}{2!}\left(-\frac{1}{\sqrt{2}}\right)+\frac{x^3}{3!}\left(-\frac{1}{\sqrt{2}}\right)+\frac{x^4}{4!}\frac{1}{\sqrt{2}}+\cdots$$

$$=\frac{1}{\sqrt{2}}[1+x-\frac{x^2}{2!}-\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots]$$

Put $x=-1°=-0.01745$

$\sin 44° = \sin(45° - 1°) = \frac{1}{\sqrt{2}}[1 + (-0.01745) - \frac{(-0.01745)^2}{2!} - \frac{(-0.01745)^3}{3!} + \frac{(-0.01745)^4}{4!} + \cdots]$

≈ 0.6945

Put $x = 1° = 0.01745$

$\sin 46° = \sin(45° + 1°) = \frac{1}{\sqrt{2}}[1 + (0.01745) - \frac{(0.01745)^2}{2!} - \frac{(0.01745)^3}{3!} + \frac{(0.01745)^4}{4!} + \cdots]$

≈ 0.7193

6 Expand $tan(x + \frac{\pi}{4})$ in powers of x. Hence find the value of tan 50°

Solution:

By Taylor's series

$$(x+h) = f(h) + xf'(h) + \frac{x^2}{2!}f''(h) + \frac{x^3}{3!}f'''(h) + \cdots + \frac{x^n}{n!}f^n(h) + \cdots$$

Suppose $\tan\left(x + \frac{\pi}{4}\right) = f(x+h) \Rightarrow f(x) = tanx$ and $h = 45° = \frac{\pi}{4}$

Now

$f(x) = \tan x \qquad \Rightarrow f(h) = f\left(\frac{\pi}{4}\right) = \tan\left(\frac{\pi}{4}\right) = 1$

$f'(x) = \sec^2 x \qquad \Rightarrow f'\left(\frac{\pi}{4}\right) = 2$

$f''(x) = 2\sec^2 x\, tanx \qquad \Rightarrow f''\left(\frac{\pi}{4}\right) = 4$

$f'''(x) = 4\sec^2 x tan^2 x + 2\sec^4 x \qquad \Rightarrow f'''\left(\frac{\pi}{4}\right) = 16$

Substituting the values in series

$$f(x+h) = f(h) + xf'(h) + \frac{x^2}{2!}f''(h) + \frac{x^3}{3!}f'''(h) + \cdots + \frac{x^n}{n!}f^n(h) + \cdots$$

$$\tan\left(x + \frac{\pi}{4}\right) = f\left(\frac{\pi}{4}\right) + xf'\left(\frac{\pi}{4}\right) + \frac{x^2}{2!}f''\left(\frac{\pi}{4}\right) + \frac{x^3}{3!}f'''\left(\frac{\pi}{4}\right) + \cdots + \frac{x^n}{n!}f^n\left(\frac{\pi}{4}\right) + \cdots$$

$$= 1 + 2x + 4\frac{x^2}{2!} + 16\frac{x^3}{3!} + \cdots$$

$$= 1 + 2x + 2x^2 + \frac{8}{3}x^3 + \cdots$$

Put $x = 5° = (0.08726)$

$$\tan 50° = \tan(45° + 5°) = 1 + 2x + 2x^2 + \frac{8}{3}x^3 + \cdots$$

$$= 1 + 2(0.08726) + 2(0.08726)^2 + \frac{8}{3}(0.08726)^3 + \cdots$$

≈ 1.19753

7 Using Taylor's series find $\sqrt[3]{27.12}$ correct to four decimal places. **(Jun-2019-OLD)**

Solution:

By Taylor's series

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \cdots + \frac{h^n}{n!}f^n(x) + \cdots$$

Suppose $f(x+h) = \sqrt[3]{x+h} \Rightarrow f(x) = \sqrt[3]{x}$

$f(x+h) = \sqrt[3]{x+h} = \sqrt[3]{27+0.12} = \sqrt[3]{27.12}$

Let $x = 27$ and $h = 0.12$

$$f(x) = x^{\frac{1}{3}} \qquad \Rightarrow f(27) = 3$$

$$f'(x) = \frac{1}{3}x^{-\frac{2}{3}} \qquad \Rightarrow f'(27) = \frac{1}{27}$$

$$f''(x) = -\frac{2}{9}x^{-\frac{5}{3}} \qquad \Rightarrow f''(27) = -\frac{2}{2187}$$

$$f'''(x) = \frac{10}{27}x^{-\frac{8}{3}} \qquad \Rightarrow f'''(27) = \frac{10}{177147}$$

Substituting the values in series

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \cdots + \frac{h^n}{n!}f^n(x) + \cdots$$

$$= 3 + (0.12)\left(\frac{1}{27}\right) + \frac{(0.12)^2}{2!}\left(-\frac{2}{2187}\right) + \frac{(0.12)^3}{3!}\frac{10}{177147} + \cdots$$

$$\approx 3.00443$$

TOPIC:13 MACLAURIN SERIES

MACLAURIN SERIES:

Statement: If ***f(x)*** is a function of ***x*** which can be expanded in to positive ascending powers of ***x*** then

$$\boldsymbol{f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^n(0) + \cdots}$$

NUMERICAL QUESTIONS

1 Find the Maclaurin's series for e^x

	Solution: By Maclaurin series $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^n(0) + \cdots$ $f(x) = e^x \Rightarrow f(0) = 1$ $f'(x) = e^x \Rightarrow f'(0) = 1$ $f''(x) = e^x \Rightarrow f''(0) = 1$ $f'''(x) = e^x \Rightarrow f'''(0) = 1$ $f''''(x) = e^x \Rightarrow f''''(0) = 1$ Substituting the values in series $e^x = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^n(0) + \cdots$ $= 1 + x1 + \frac{x^2}{2!}1 + \frac{x^3}{3!}1 + \cdots + \frac{x^n}{n!}1 + \cdots$ $\boldsymbol{e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots}$ ❖ Put $x = -x$ we get $\boldsymbol{e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + - \cdots}$
2	Write down Maclaurin series for $\sin x$ **Solution:** By Maclaurin series $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^n(0) + \cdots$ $f(x) = \sin x \Rightarrow f(0) = 0$ $f'(x) = \cos x \Rightarrow f'(0) = 1$ $f''(x) = -\sin x \Rightarrow f''(0) = 0$ $f'''(x) = -\cos x \Rightarrow f'''^{(0)} = -1$ $f''''^{(x)} = \sin x \Rightarrow f''''^{(0)} = 0$ Substituting the values in series $\sin x = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^n(0) + \cdots$

	$= 0 + x1 + \frac{x^2}{2!}0 + \frac{x^3}{3!}(-1) + \frac{x^4}{4!}0 + \frac{x^5}{5!}1 - + \cdots$ $= x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$
3	Write down Maclaurin series for $\sin hx$ **Solution:** By Maclaurin series $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^n(0) + \cdots$ $f(x) = \sinh x \implies f(0) = 0$ $f'(x) = \cos hx \implies f'(0) = 1$ $f''(x) = \sinh x \implies f''(0) = 0$ $f'''(x) = \cos hx \implies f'''^{(0)} = 1$ $f''''^{(x)} = \sinh x \implies f''''^{(0)} = 0$ Substituting the values in series $\sinh x = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^n(0) + \cdots$ $= 0 + x1 + \frac{x^2}{2!}0 + \frac{x^3}{3!}(1) + \frac{x^4}{4!}0 + \frac{x^5}{5!}1 + \cdots$ $\boldsymbol{\sin hx = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots}$
4	Write down Maclaurin series for $\cos x$ **Solution:** By Maclaurin series $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^n(0) + \cdots$ $f(x) = \cos x \implies f(0) = 1$ $f'(x) = -\sin x \implies f'(0) = 0$ $f''(x) = -\cos x \implies f''(0) = -1$ $f'''(x) = \sin x \implies f'''(0) = 0$ $f''''(x) = \cos x \implies f''''(0) = 1$ Substituting the values in series

	$\cos x = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^n(0) + \cdots$ $= 1 + x0 + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(0) + \frac{x^4}{4!}1 + \frac{x^5}{5!}0 - + \cdots$ $\boldsymbol{\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots}$
5	Write down Maclaurin series for $\cos hx$ **Solution:** By Maclaurin series $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^n(0) + \cdots$ $f(x) = \cos hx \quad \Rightarrow f(0) = 1$ $f'(x) = \sin hx \quad \Rightarrow f'(0) = 0$ $f''(x) = \cos hx \quad \Rightarrow f''(0) = 1$ $f'''(x) = \sin hx \quad \Rightarrow f'''(0) = 0$ $f''''(x) = \cos hx \quad \Rightarrow f''''(0) = 1$ Substituting the values in series $\cos hx = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^n(0) + \cdots$ $= 1 + x0 + \frac{x^2}{2!}(1) + \frac{x^3}{3!}(0) + \frac{x^4}{4!}1 + \frac{x^5}{5!}0 - + \cdots$ $\boldsymbol{\cos hx = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots}$
6	Obtain the Maclaurin's series of $log_e(1+x)$ **Solution:** By Maclaurin series $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^n(0) + \cdots$ $f(x) = \log(1+x) \quad \Rightarrow f(0) = 0$ $f'(x) = \frac{1}{1+x} \quad \Rightarrow f'(0) = 1$ $f''(x) = -\frac{1}{(1+x)^2} \quad \Rightarrow f''(0) = -1$

	$f'''(x) = \frac{2}{(1+x)^3} \quad \Rightarrow \quad f'''(0) = 2$ $f''''(x) = -\frac{6}{(1+x)^4} \quad \Rightarrow f''''(0) = -6$ Substituting the values in series $log_e(1+x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^n(0) + \cdots$ $= 0 + x1 + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(2) + \frac{x^4}{4!}(-6) + \cdots$ $= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$
7	Expand $e^{\sin x}$ by Maclaurin's series up to the terms containing x^4 **Solution:** Let $sin\ x = t$ $e^{sinx} = e^t = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \cdots$ Put $t = sinx = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$ $e^{sinx} = 1 + \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots\right) + \frac{(x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots)^2}{2!} + \frac{(x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots)^3}{3!} + \cdots$ $= 1 + x + \frac{x^2}{2} - \frac{x^4}{8} + \cdots$
8	Expand $\tan^{-1} x$ in power of x **Solution:** Let $y = \tan^{-1} x$ $\frac{dy}{dx} = \frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \cdots$ Integrating we get $y = \tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots + c$ Putting $x = 0 \Rightarrow y = \tan^{-1} 0 = 0$ $c = 0$ Hence $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$

9 Expand $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ in powers of x, using Maclaurin's Series. **OR**

Prove that $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right) = \frac{1}{2}(x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots)$

Solution:

Let $x = tan\,\theta \implies \theta = tan^{-1}x$

$$\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right) = \tan^{-1}\left(\frac{\sqrt{1+\tan^2\theta}-1}{\tan\theta}\right)$$

$$= \tan^{-1}\left(\frac{\sec\theta - 1}{\tan\theta}\right)$$

$$= \tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right)$$

$$= \tan^{-1}\left(\frac{2\sin^2\frac{\theta}{2}}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}\right)$$

$$= \tan^{-1}\left(\tan\frac{\theta}{2}\right)$$

$$= \frac{\theta}{2} = \frac{\tan^{-1}x}{2}$$

$$\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right) = \frac{1}{2}(x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots)$$

Extreme Values of Functions

Let f be a function defined over an interval I and let c ∈ I. We say f has an **absolute maximum** on I at c if f(c) ≥ f(x) for all x ∈ I

Let f be a function defined over an interval I and let c ∈ I. We say f has an **absolute minimum** on I at c if f(c) ≤ f(x) for all x ∈ I

An extremum (or extreme value) of a function is a point at which a maximum or minimum value of the function is obtained in some interval. A local extremum (or relative extremum) of a function is the point at which a maximum or minimum value of the function in some open interval containing the point is obtained

Let c be an interior point in the domain of f. We say that c is a **critical point** of f if $f'(c) = 0$ *or* $f'(c)$ is undefined

Absolute extrema may be determined according to the following method

Given a function f and interval $[a, b]$

1Determine all critical points of f in the interval $[a, b]$

2 Determine the value of f at each of its critical points

	3 Determine the value of f at each of the endpoints. The point(s) corresponding to the largest values of f are the absolute maximum (maxima), and the point(s) corresponding to the smallest values of f are the absolute minimum (minima). The other values may be local extrema
	FIRST DERIVETIVE TEST Let f be the function defined in an open interval I. And f be continuous at critical point c in I such that f'(c) = 0. 1. If f'(x) changes sign from positive to negative as x increases through point c, then c is the point of local maxima, and the f(c) is the maximum value. 2. If f'(x) changes sign from negative to positive as x increases through point c, then c is the point of local minima, and the f(c) is the minimum value. 3. If f'(x) doesn't change sign as x increases through c, then c is neither a point of local nor a point of local maxima. It will be called the point of inflection. **Second Derivative Test** Let f be the function defined on an interval I, and it is two times differentiable at c. i. x = c will be point of local maxima if f'(c) = 0 and f''(c)<0. Then, f(c) will be having local maximum value. ii. x = c will be point of local minima if f'(c) = 0 and f''(c) > 0. Then, f(c) will be having local minimum value. iii. When both f'(c) = 0 and f''(c) = 0, the test fails, and the first derivative test will give you the value of local maxima and minima
1	**Find absolute maximum and minimum value of function $2x^3 + 3x^2 - 12x$ on interval [-1,2]** **Solution:** Step 1: Find the derivative of f(x) $f'(x) = 6x^2 + 6x - 12$ Step 2: Equate $f'(x) = 0$ to get the critical numbers. We achieve this by factoring the polynomials $f'(x) = 6(x-1)(x+2) = 0$ $x = 1, -2$ Only x=1 belongs to the interval[-1,2] The value of f at each of its critical points and end points $f(1) = 2 \cdot 1^3 + 3 \cdot 1^2 - 12 \cdot 1 = -7$ $f(-1) = 2 \cdot -1^3 + 3 \cdot -1^2 - 12 \cdot -1 = 13$

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$f(2) = 2 \cdot 2^3 + 3 \cdot 2^2 - 12 \cdot 2 = 4$

Maximum value of function is 13 at $x = -1$

Minimum value of function is -7 at $x = 1$

2 **Find where the function $f(x) = x^3 - 12x - 5$ is increasing and where it is decreasing**

Solution:

$f(x) = x^3 - 12x - 5$

Step 1: Find the derivative of f(x)

$f'(x) = 3x^2 - 12 = 3(x^2 - 4) = 3(x - 2)(x + 2)$

Step 2: Equate f '(x) = 0 to get the critical numbers. We achieve this by factoring the polynomials.

$f'(x) = 3(x - 2)(x + 2) = 0$

x = 2, –2

Step 3: Determine the sign of f '(x) for each interval by setting up intervals with the essential integers as their endpoints. For each interval, use the increasing/decreasing test to determine if f(x) is increasing or decreasing.

Interval	x – 2	x + 2	f'(x)	f
x < – 2	–	–	+	Increasing on $(-\infty, -2)$
–2 < x < 2	–	+	–	Decreasing on (–2, 2)
x > 2	+	+	+	Increasing on $(2, \infty)$

3 **Find extreme value of function $f(x) = x^3 - 12x - 5$**

Solution:

$f(x) = x^3 - 12x - 5$

Step 1: Find the derivative of f(x)

$f'(x) = 3x^2 - 12 = 3(x^2 - 4) = 3(x - 2)(x + 2)$

Step 2: Equate f '(x) = 0 to get the critical numbers. We achieve this by factoring the polynomials.

$f'(x) = 3(x - 2)(x + 2) = 0$

x = 2, –2

Step 3: Determine the sign of f '(x) for each interval by setting up intervals with the essential integers as their endpoints. For each interval, use the increasing/decreasing test to determine if f(x) is increasing or decreasing.

Interval	x – 2	x + 2	f'(x)	f

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x < – 2	–	–	+	Increasing on $(-\infty, -2)$
–2 < x < 2	–	+	–	Decreasing on (–2, 2)
x > 2	+	+	+	Increasing on $(2, \infty)$

At $x = -2$ $f'(x)$ changes sign from positive to negative so $x = -2$ is the point of local maxima, and the $f(-2) = 11$ is the maximum value

At $x = 2$ $f'(x)$ changes sign negative to positive so $x = 2$ is the point of local minima and $f(2) = -21$ is minimum value.

4 **Find the value of local maximum and local minimum of $f(x) = x^3 - 9x^2 + 24x + 2$**

Solution:

Critical point $f'(x) = 0$

$f'(x) = 3x^2 - 18x + 24 = 0$

$3(x-2)(x-4) = 0$

$x = 2$ *and* $x = 4$ are critical points.

Now $f''(x) = 6x - 18$

The value of $f''(x)$ at critical points

$f''(2) = -6 < 0 \Longrightarrow$local maximum

$f''(4) = 6 > 0 \Longrightarrow$local minimum

$f(2) = 22$ is local maximum value

$f(4) = 18$ is local minimum value.

5 **Find the maxima and the minima of the function $f(x) = x^3 - 12x + 5$, by using the second derivative test**

Solution:

The given function is $f(x) = x^3 - 12x + 5$

$f'(x) = 3x^2 - 12$

f'(x) = 0, Hence we have x = 2, and x = -2

f''(x) = 6x

f''(2) = 6 x 2 = 12 , and f''(2) > 0 , and x = 2 is the minima

f''(-2) = 6 x (-2) = -12, and f''(-2) < 0, and x = -2 is the maxima.

Therefore by using the second derivative test, the local maxima is -2, with a maximum value of f(-2) = 21, and the local minima is 2, with a minimum value of f(2) = -11.